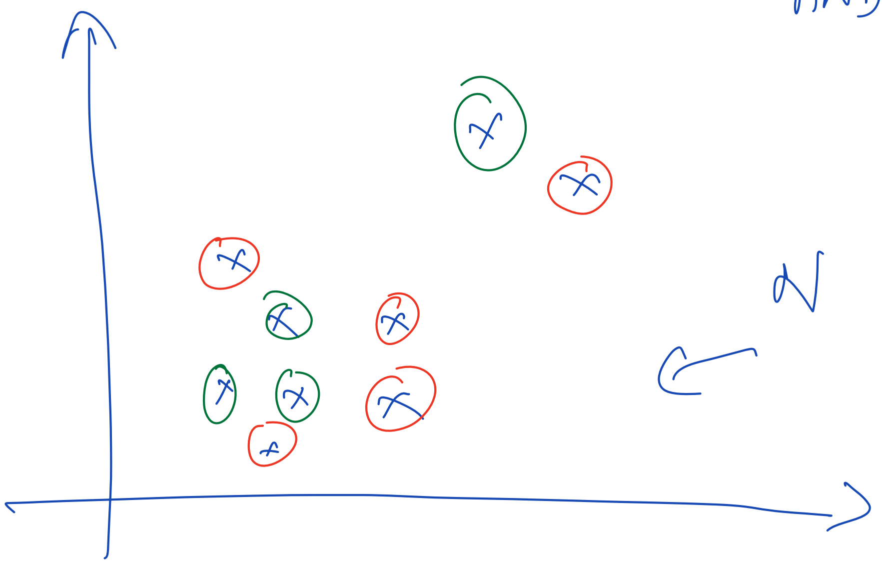
# Last Class (Oct 13)

1) Case study : customer grouping
2) DBSCAN : key ideas
3) DBSCAN Algorithm
4) Adjusting params
5) Advantages & Disadvantages of DBSCAN
6) DBSCAN implementation
7) Intro to Anomaly / Novelty / Outlier Detection
8) ~~Business Case Intro~~
9) ~~RANSAC & Elliptical Envelope~~

# Today's class

1) Business Case Intro
2) RANSAC & Elliptical Envelope
3) Isolation Forests
4) One-class SVM (very brief)
5) Local Outlier Factor
6) Comparison of all methods

# RANSAC → Random Sampling AND Consensus



$N$ $\quad n_1$

| # of try | inliers | outliers |
|---|---|---|
| 1 | $x_1, x_2, x_{10}, \ldots$ | $x_5, x_6, \ldots$ |
| 2 | $x_{10}, x_{11}, \ldots$ | $x_1, x_2, \ldots$ |
| ⋮ | | |
| 50 | | |

$x_1$ → # of times it is outlier → 10

\# of times it is inlier → $\boxed{40}$ → 50

max

outlier

$x_2$ → $\boxed{14\ \cancel{5}}$ 5 → inlier

$x_3$ → $\boxed{30}$, 20 → max times → outlier

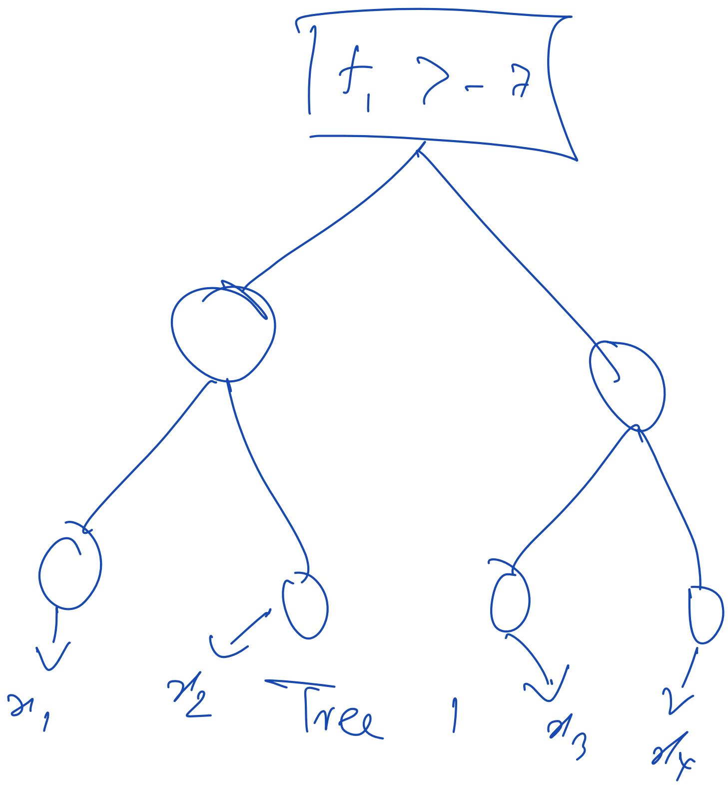
max times → outlier

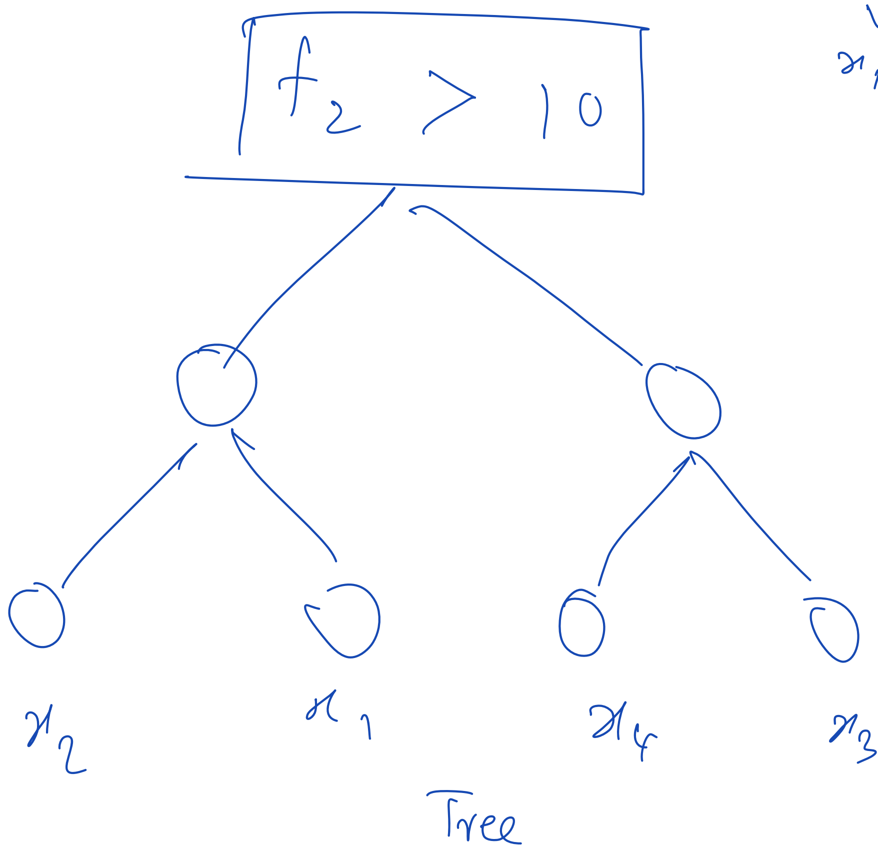
$x_1 \rightarrow$ inlier

$x_2, x_3 \rightarrow$ outlier

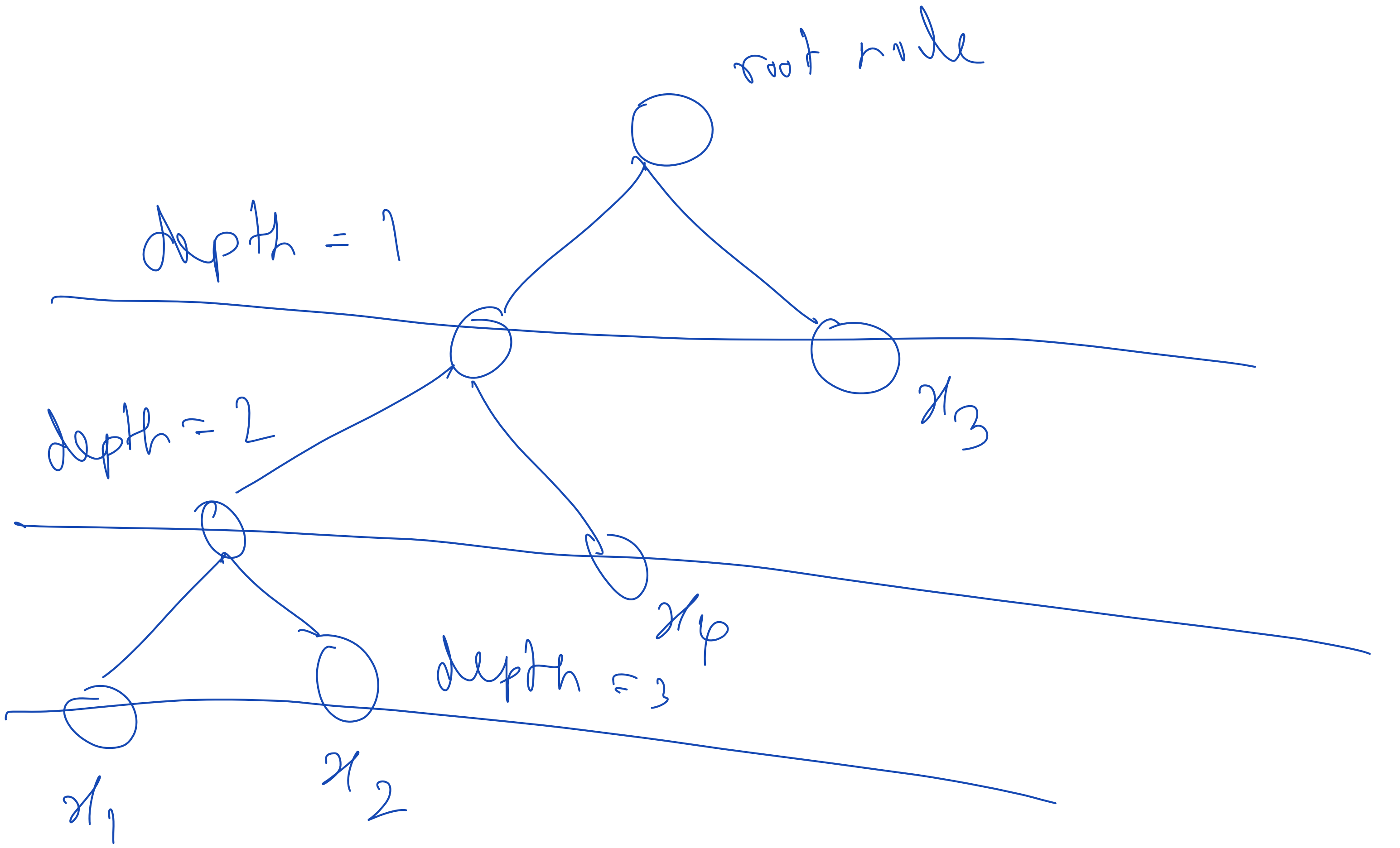
| points | $f_1$ | $f_2$ |
|---|---|---|
| $x_1$ | 3 | 4 |
| $x_2$ | -7 | 10 |
| $x_3$ | 10 | -1 |
| $x_4$ | 2 | 3 |

$f_1$




Ultimate goal: every leaf node has 1 data-point




**Single Tree**

$x_1, x_2$ : depth 3

$x_4$ : depth 2

$x_3$ : depth 1

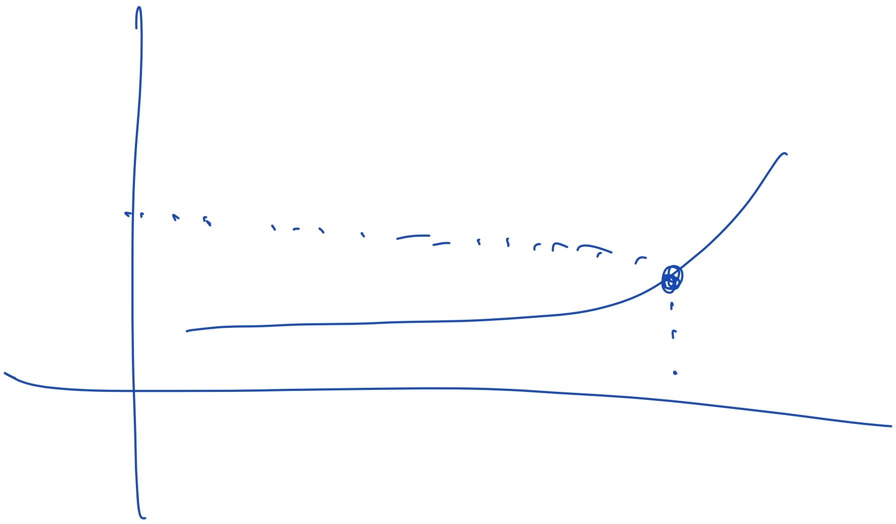
| | Tree 1 | Tree 2 | Tree 3 | |
|---|---|---|---|---|
| $x_1$ : | depth 4 | depth 5 | depth 1 | ... |

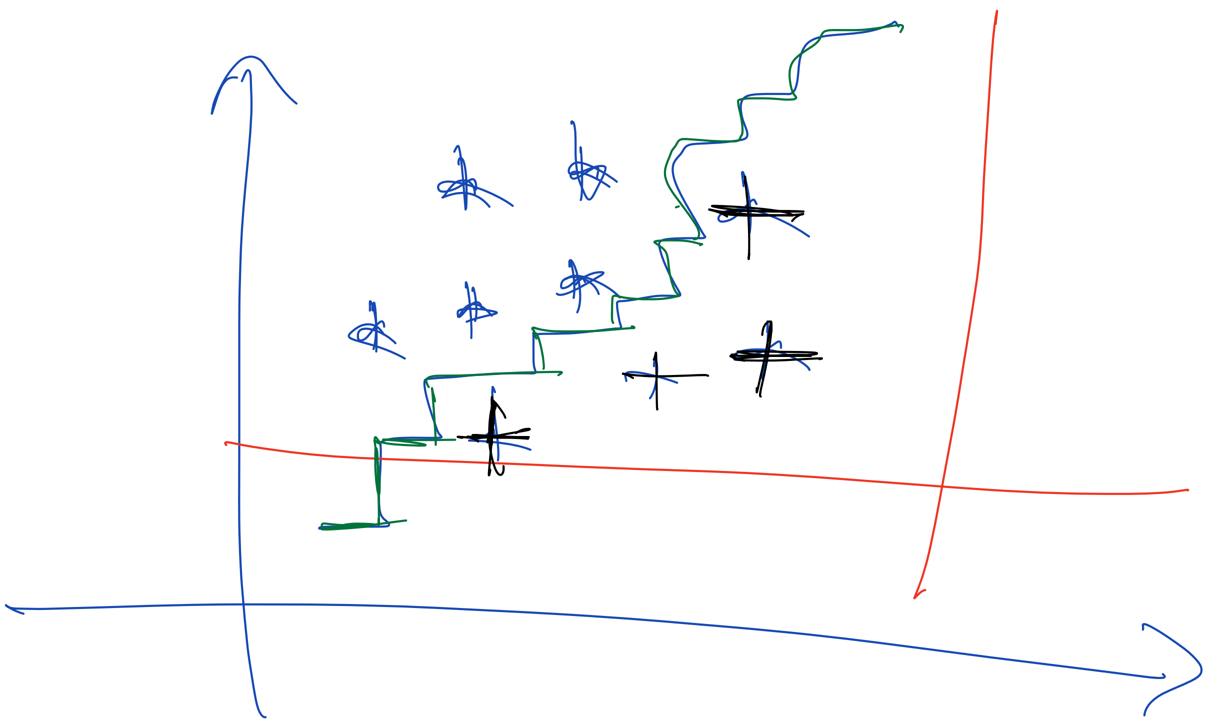
100 trees : $\longrightarrow$ $x_1 \longrightarrow$ average depth $(d_1)$

$x_2 \longrightarrow d_2$

$x_3 \longrightarrow d_3$

$x_4 \longrightarrow d_4$





$\boxed{A}$

| $B_1$ | $B_2$ | $B_3$ | $B_4$ | $B_5$ | $B_6$ |
|---|---|---|---|---|---|
| <u>7.5</u> | <u>8.5</u> | <u>9.5</u> | 6.5 | 5.5 | 4.5 |

$4.5 \to B_6$ (1)

$5.5 \to B_5$ (2)

$6.5 \to B_4$ (3)

$7.5 \to B_1$ (4)

$\boxed{8.5 \to B_2}$ (5)

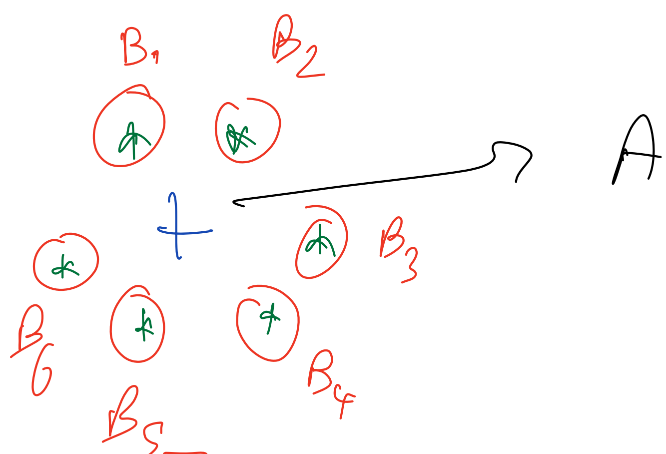
$9.5 \to B_3$ (6)

$k = 5$

5th nn

k-distance of $A \to 8.5$

$\{B_6, B_5, B_4, B_1, B_2\} \to \boxed{N_5(A)}$




LRD (A)

avg of LRD of $[B_1\ B_2\ B_3\ B_4\ B_5]$

LRD (A)